

ReCodeAgent paper Tables 1 and 2 with CodeWeaver results

Paper values: arXiv:2604.07341 and the official results.xlsx (MD5 a404779f2dcd7ac44d43bf72f4e88b98). CodeWeaver values: variant='full', repetition=0. P-Tool is the paper's comparison tool; P-RCA is the paper's ReCodeAgent result; CW is independently measured CodeWeaver. A trailing * marks a partial measured sum/mean where one or more constituent CodeWeaver cells were unavailable. '-' means unavailable or not applicable and is never converted to zero.

Table 1 - Effectiveness

Effectiveness of ReCodeAgent in repository-level code translation and validation in terms of test and function validation. Tuple components from the paper are expanded into P-Tool and P-RCA columns; CodeWeaver is placed immediately beside the corresponding paper metric. E=expected, TE=executed, TP=passing, TF=failing, C=coverage before generated tests, C+=coverage after generated tests, T=total, S=success, F=fail.

Tool (PL)	Project	LoC	CS % P-Tool	CS % P-RCA	CS % CW	Validated developer tests E/TE/TP/TF P-Tool	Validated developer tests E/TE/TP/TF P-RCA	Validated developer tests E/TE/TP/TF CW	Translated developer tests TE/TP/TF P-RCA	Translated developer tests TE/TP/TF CW	Generated tests TE/TP/TF P-RCA	Generated tests TE/TP/TF CW	Coverage C/C+ % P-RCA	Coverage C/C+ % CW	Function validation T/S/F P-Tool	Function validation T/S/F P-RCA	Function validation E/TE/S/F CW
Oxidizer (Go->Rust)	checkdigit	428	100	100	100	36/36/33/3	36/36/36/0	36/26/26/0	36/36/0	55/55/0	71/71/0	19/19/0	79.7/94.7	63.5/92.5	29/21/8	29/29/0	29/12/12/0
Oxidizer (Go->Rust)	go-edlib	639	100	100	100	36/36/19/17	36/36/36/0	36/2/2/0	36/36/0	77/77/0	3/3/0	35/35/0	94.7/94.9	7.1/40.2	24/18/6	24/24/0	24/7/7/0
Oxidizer (Go->Rust)	histogram	314	100	100	100	2/2/2/0	2/2/2/0	2/3/3/0	2/2/0	20/20/0	66/66/0	18/18/0	38.0/90.5	30.6/96.6	19/12/7	19/19/0	19/-/-/-
Oxidizer (Go->Rust)	nameparts	413	100	100	100	26/26/23/3	26/26/26/0	26/-/-/-	26/26/0	48/48/0	22/22/0	22/22/0	96.8/96.8	-	15/9/6	15/14/1	15/-/-/-
Oxidizer (Go->Rust)	stats	1,241	100	100	100	121/121/71/50	121/121/121/0	121/0/0/0	121/121/0	189/189/0	320/320/0	67/67/0	43.3/79.6	0/81.6	52/38/14	52/52/0	52/1/1/0
Oxidizer (Go->Rust)	textrank	1,132	100	100	100	8/8/6/2	8/8/8/0	8/3/3/0	8/8/0	37/37/0	127/127/0	28/28/0	72.6/98.7	20.3/95.1	52/40/12	52/52/0	52/-/-/-
Oxidizer (Go->Rust)	Total	4,167	100	100	100	229/229/154/75	229/229/229/0	229/34*/34*/0*	229/229/0	426/426/0	609/609/0	189/189/0	70.8/92.5	24.3*/81.2*	191/138/53	191/190/1	191/20*/20*/0*
AlphaTrans (Java->Python)	cli	37,841	100	100	100	381/66/35/31	381/381/360/21	381/381/370/11	381/381/0	-	257/257/0	46/46/0	96.7/97.3	90.7/94.6	257/196/61	257/241/16	257/-/-/-
AlphaTrans (Java->Python)	csv	33,072	100	100	100	298/147/3/144	298/298/241/57	298/-/-/-	298/298/0	-	192/190/2	398/398/0	84.4/85.8	38.3/79.6	213/74/139	213/211/2	213/-/-/-
AlphaTrans (Java->Python)	fileupload	3,567	100	100	100	39/39/36/3	39/39/39/0	39/9/9/0	39/39/0	-	208/208/0	45/45/0	38.7/71.6	46.7/93.7	25/19/6	25/25/0	25/-/-/-
AlphaTrans (Java->Python)	validator	41,605	100	100	100	463/359/114/245	463/463/438/25	463/8/0/8	463/435/28	-	132/131/1	105/105/0	65.0/73.0	40.5/78.4	409/217/192	409/397/12	409/-/-/-
AlphaTrans (Java->Python)	Total	116,085	100	100	100	1,181/611/188/423	1,181/1,181/1,078/103	1,181/398*/379*/19*	1,181/1,153/28	-	789/786/3	594/594/0	71.2/81.9	54.1/86.6	904/506/398	904/874/30	904/-/-/-
SKEL (Python->JavaScript)	bst	123	100	100	100	11/11/11/0	11/11/11/0	11/11/11/0	11/11/0	-	6/6/0	0/0/0	89.7/99.0	54.6/54.6	21/21/0	21/21/0	21/-/-/-
SKEL (Python->JavaScript)	colorsys	120	100	100	100	2/2/2/0	2/2/2/0	2/-/-/-	2/2/0	-	46/46/0	0/0/0	87.0/91.3	-	9/9/0	9/9/0	9/-/-/-
SKEL (Python->JavaScript)	heapq	189	100	100	100	8/8/7/1	8/8/8/0	8/8/8/0	8/8/0	-	11/11/0	6/6/0	91.6/91.6	76.0/95.7	24/23/1	24/24/0	24/-/-/-
SKEL (Python->JavaScript)	html	684	100	100	100	7/7/6/1	7/7/7/0	7/7/7/0	7/7/0	-	13/13/0	0/0/0	77.1/86.1	71.1/71.1	42/39/3	42/42/0	42/-/-/-
SKEL (Python->JavaScript)	mathgen	735	100	100	100	5/5/4/1	5/5/5/0	5/5/0/5	5/5/0	-	11/11/0	0/0/0	96.4/98.6	23.4/23.4	82/79/3	82/82/0	82/-/-/-
SKEL (Python->JavaScript)	rbt	366	100	100	100	10/10/10/0	10/10/10/0	10/10/10/0	10/10/0	-	5/5/0	0/0/0	87.0/88.3	63.9/63.9	27/27/0	27/27/0	27/-/-/-
SKEL (Python->JavaScript)	strsim	654	100	100	100	19/19/19/0	19/19/19/0	19/19/0/19	19/19/0	-	64/64/0	0/0/0	88.8/94.1	47.6/47.6	50/50/0	50/50/0	50/-/-/-
SKEL (Python->JavaScript)	toml	1,206	100	100	100	12/12/10/2	12/12/12/0	12/12/4/8	12/12/0	-	150/150/0	0/0/0	72.6/83.2	39.0/39.0	47/43/4	47/47/0	47/-/-/-
SKEL (Python->JavaScript)	Total	4,077	100	100	100	74/74/69/5	74/74/74/0	74/72*/40*/32*	74/74/0	-	306/306/0	6/6/0	86.3/91.5	53.7*/56.5*	302/291/11	302/302/0	302/-/-/-
SWE-agent (C->Rust)	Crust- α	22,961	40	40	100	166/153/130/23	166/166/146/20	166/150*/136*/14*	-	1,158/1,151/7	493/493/0	953/951/2	68.2/75.7	64.2*/91.2*	673/-/-	673/-/-	-
SWE-agent (C->Rust)	Crust- β	66,704	0	49	100	321/0/0/0	321/320/295/25	321/337*/304*/33*	-	1,723*/1,699*/24*	1,118/1,114/4	1,432/1,431/1	57.9/75.2	51.1/86.8	1,900/-/-	1,900/-/-	-
SWE-agent (C->Rust)	Crust- σ	3,894	1	0	100	1/1/1/0	1/0/0/0	1/-/-/-	-	66/66/0	53/51/2	53/53/0	4.3/67.4	44.8/69.3	41/-/-	41/-/-	-
SWE-agent (C->Rust)	Crust- γ	15,169	0	0	100	135/0/0/0	135/0/0/0	135/68*/60*/8*	-	296*/288*/8*	274/270/4	219/217/2	27.0/44.7	54.1*/75.4*	572/-/-	572/-/-	-
SWE-agent (C->Rust)	Total	108,728	41	89	100	623/154/131/23	623/486/441/45	623/555*/500*/55*	-	3,243*/3,204*/39*	1,938/1,928/10	2,657/2,652/5	39.3/65.7	53.5*/80.6*	3,186/-/-	3,186/-/-	-
Total	Total	233,057	96.9	99.4	100	2,107/1,068/542/526	2,107/1,970/1,822/148	2,107/1,059*/953*/106*	1,484/1,456/28	3,669*/3,630*/39*	3,642/3,629/13	3,446/3,441/5	70.8/85.4	46.4*/73.5*	4,583/935/462	4,583/1,366/31	1,397/20*/20*/0*

Table 2 - Test Translation

Comparison between ReCodeAgent translated tests and original source-language tests, with the same official AST/runtime protocol applied to CodeWeaver. Paper subtotal means follow the workbook's SUBTOTAL(AVERAGE) formulas. M/N=matching/non-matching, T/M=total/matching, S/T=source/target.

Tool (PL)	Project	Tests Paper	Tests CW	Translated/Not Paper	Translated/Not CW	Assertion count M/N Paper	Assertion count M/N CW	assertEqual output T/M Paper	assertEqual output T/M CW	Type match Equal % Paper	Type match Equal % CW	Type match True % Paper	Type match True % CW	Type match False % Paper	Type match False % CW	Type match Other % Paper	Type match Other % CW	Avg cosine Paper	Avg cosine CW	Avg LoC S/T Paper	Avg LoC S/T CW	Avg calls S/T Paper	Avg calls S/T CW
Oxidizer (Go->Rust)	checkdigit	36	36	36/0	36/0	36/0	12/24	45/45	0/0	100	40	-	-	-	-	-	-	0.94	0.86	24.42/24.58	24.42/8.25	7.14/5.33	7.14/3.67
Oxidizer (Go->Rust)	go-edlib	36	36	36/0	36/0	36/0	19/17	45/45	0/0	84.44	91.11	-	-	-	-	-	-	0.92	0.83	30.78/51	30.78/16.44	7.61/9.50	7.61/4.72
Oxidizer (Go->Rust)	histogram	2	2	2/0	2/0	2/0	2/0	11/11	0/0	100	100	-	-	-	-	-	-	0.96	0.85	23.50/16.50	23.50/34	24/16.50	24/17.50
Oxidizer (Go->Rust)	nameparts	26	26	26/0	24/2	26/0	14/12	51/51	0/0	100	90.20	-	-	-	-	-	-	0.94	0.87	11.96/6.31	11.96/5.71	6.15/3.81	6.12/2.54
Oxidizer (Go->Rust)	stats	121	121	121/0	121/0	121/0	58/63	150/150	0/0	91.15	84.96	-	-	-	-	-	-	0.85	0.85	16.82/9.44	14.16/11.45	8.12/6	7.62/7.14
Oxidizer (Go->Rust)	textrank	8	8	8/0	8/0	8/0	2/6	12/12	0/0	100	-	-	-	-	-	-	-	0.91	0.87	13.75/15.25	13.75/11.75	10.88/11.88	10.88/9.25
Oxidizer (Go->Rust)	Total	229	229	229/0	227/2	229/0	107/122	314/314	0/0	95.93	81.25	-	-	-	-	-	-	0.92	0.85	20.21/20.51	19.76/14.60	10.65/8.84	10.56/7.47
AlphaTrans (Java->Python)	cli	381	381	381/0	257/124	381/0	199/182	452/452	196/190	99.61	79.57	99.68	87.78	98.37	79.41	97.87	47.73	0.90	0.75	12.41/11.59	13.03/11.97	13.30/12.63	13.06/9.18
AlphaTrans (Java->Python)	csv	298	298	298/0	298/0	292/6	242/56	207/207	149/118	100	67.30	81.82	36.73	84.31	18.65	92.86	35.29	0.90	0.75	11.84/8.94	11.84/6.61	12.20/10.46	12.20/5.67
AlphaTrans (Java->Python)	fileupload	39	39	39/0	39/0	39/0	37/2	37/37	14/14	100	78.38	100	16.67	80	20	100	50	0.87	0.71	6.74/7.38	6.74/5.49	5.87/6.36	5.87/3.49
AlphaTrans (Java->Python)	validator	463	463	463/0	296/167	458/5	171/292	374/374	122/120	98.52	66.04	99.28	38.99	99.49	24.14	90.72	13.23	0.89	0.79	17.69/17.23	18.70/12.18	18.78/18.30	21.19/7.48
AlphaTrans (Java->Python)	Total	1,181	1,181	1,181/0	890/291	1,170/11	649/532	1,070/1,070	481/442	99.53	72.82	95.19	45.04	90.54	35.55	95.36	36.56	0.89	0.75	12.17/11.29	12.58/9.06	12.54/11.94	13.08/6.46
SKEL (Python->J avaScript)	bst	11	11	11/0	0/11	11/0	0/11	74/74	0/0	100	-	-	-	-	-	-	-	0.91	-	22.27/22.64	-	5.73/12	-
SKEL (Python->J avaScript)	colorsys	2	2	2/0	2/0	2/0	0/2	39/39	0/0	100	-	-	-	-	-	-	-	0.93	0.87	67/65	67/138	45/44	45/7
SKEL (Python->J avaScript)	heapq	8	8	8/0	8/0	8/0	6/2	29/29	0/0	100	-	-	-	-	-	-	-	0.90	0.89	13.12/13.38	13.12/12.25	11.25/10.25	11.25/10.12
SKEL (Python->J avaScript)	html	7	7	7/0	7/0	7/0	6/1	19/19	0/0	100	-	-	-	-	-	-	-	0.92	0.91	24.71/22.71	24.71/20	20.29/19.29	20.29/16.86
SKEL (Python->J avaScript)	mathgen	5	5	5/0	5/0	5/0	0/5	163/163	0/0	100	-	-	-	-	-	-	-	0.93	0.90	47/51	47/75	49/42	49/44
SKEL (Python->J avaScript)	rbt	10	10	10/0	10/0	10/0	8/2	16/16	0/0	100	-	-	-	-	-	-	-	0.91	0.90	17.90/19.60	17.90/17.60	20.90/17.10	20.90/13.90
SKEL (Python->J avaScript)	strsim	19	19	19/0	0/19	19/0	0/19	150/150	0/0	100	-	-	-	-	-	-	-	0.93	-	18.74/17.21	-	22.84/21.84	-
SKEL (Python->J avaScript)	toml	12	12	12/0	1/11	12/0	0/12	17/17	0/0	100	-	-	-	-	-	-	-	0.90	0.50	8.67/8.75	10/9	6.08/7	7/8
SKEL (Python->J avaScript)	Total	74	74	74/0	33/41	74/0	20/54	507/507	0/0	100	-	-	-	-	-	-	-	0.92	0.83	27.43/27.54	29.95/45.31	22.64/21.68	25.57/16.65
Total	Total	1,484	1,484	1,484/0	1,150/334	1,473/11	776/708	1,891/1,891	481/442	98.54	77.51	95.19	45.04	90.54	35.55	95.36	36.56	0.91	0.82	21.63/21.58	21.79/24.73	16.40/15.24	16.82/10.66